

IMEN266 Operations Research II — Homework 4 (Ch. 7, renewal theory)

Fall 2026

Not collected, not graded — but: **some Part-A problems may reappear on the exams.** Work through it with the companion notebook `HW4.ipynb`; Part-A solutions follow on PLMS.

AI rules for this homework (details: `docs/ai_usage_guide.pdf`)

- **Part A:** solve *alone first*, closed-book style, on paper. Only afterwards may you check with an AI — and if the AI changes your mind, record why in Part D.
- **Parts B–C:** AI collaboration is expected. You remain responsible for every claim you make.
- **Part D:** keep the prompt log whenever AI was used — it is your own audit trail (the homework is not collected).

Part A — Exam-style warm-up (no AI on first attempt)

- A1. (**Renewal function, limit theorems.**) Items arrive at an inspection station one at a time; the inter-arrival times are iid $\text{Erlang}(2, \lambda)$ with $\lambda = 4$ per hour (each gap is the sum of two independent $\exp(4)$ hours; mean 30 minutes). Let $N(t)$ count the arrivals in $(0, t]$ hours. (i) Give the distribution of S_n and write $P(N(t) = n)$ as a difference of two Gamma cdfs. Then use “every second event of a $\text{PP}(\lambda)$ ” to express $P(N(1) = 2)$ with Poisson probabilities and evaluate it. (ii) Show that $m(t) = \frac{\lambda t}{2} - \frac{1 - e^{-2\lambda t}}{4}$ (hint: $N(t) = \lfloor M(t)/2 \rfloor$ for the Poisson count M , and $P(M(t) \text{ odd}) = (1 - e^{-2\lambda t})/2$; or transform the renewal equation with $\tilde{F}(s) = (\lambda/(s + \lambda))^2$). Evaluate $m(1)$. (iii) State $\lim_{t \rightarrow \infty} m(t)/t$ and $\lim_{t \rightarrow \infty} (m(t) - t/\mu)$, and check the latter against $(\sigma^2 - \mu^2)/(2\mu^2)$. (iv) After the station has run for a long time an auditor arrives at an arbitrary moment. What is the expected time until the next item, and why is it not 15 minutes?
- A2. (**Stopping times and Wald’s equation.**) An inspector tests items one by one. Items are defective independently with probability $p = 0.1$; test durations $T_1, T_2, \dots$ are iid with mean 3 minutes and independent of the defect indicators $D_1, D_2, \dots$ ($D_i = 1$ if item i is defective). (i) She stops at the first defective item; N = number of items tested. Is N a stopping time for the sequence (D_i) ? For the sequence (T_i) ? Compute $E[N]$ and, with Wald’s equation, the expected total testing time. (ii) She stops after the 10th item or at the first defective, whichever comes first. Compute $E[N]$ and the expected total testing time. (iii) A colleague proposes to count only the items *before* the first defective, $N' = N - 1$. Is N' a stopping time for (D_i) ? Does $E[\sum_{i \leq N'} D_i] = E[N']E[D_1]$ hold? Does $E[\sum_{i \leq N'} T_i] = E[N']E[T_1]$ hold? Justify each answer (the three answers are not all the same). (iv) Items arrive as in A1 (gaps of mean $\mu = \frac{1}{2}$ h). Using $E[S_{N(t)+1}] = \mu(m(t) + 1)$ and A1(ii), compute the expected arrival time of the first item after $t = 10$ hours, and compare $E[S_{N(10)+1}] - 10$ with your answer to A1(iv).
- A3. (**Renewal reward: an age replacement policy.**) A component has lifetime $X \sim \text{uniform}(0, 2)$ years. It is replaced when it fails (cost $c_f = 5$) or preventively when it reaches age $T \in (0, 2]$ (cost $c_p = 1$), whichever comes first; replacements are instantaneous. (i) Identify the cycle and show $E[\text{cycle}] = T - T^2/4$ and $E[\text{cost per cycle}] = 1 + 2T$. (ii) Write the long-run cost rate

$C(T)$ and find the optimal age T^* and $C(T^*)$. Compare with the “replace only at failure” policy. (iii) Under T^* : what fraction of replacements are preventive, and what is the long-run rate of failures per year? (iv) Explain in two sentences why $C(T)$ is $E[\text{cost}]/E[\text{cycle}]$ and not $E[\text{cost}/\text{cycle length}]$.

- A4. (**Alternating renewal process.**) A web server alternates between up periods $Z \sim \text{uniform}(10, 20)$ hours and down periods. A down period consists of a deterministic 1-hour wait for the technician followed by a repair of exp duration with mean 2 hours; all durations are independent. (i) Long-run fraction of time the server is up. (ii) Long-run rate of failures (per hour). (iii) (**final-style**) Given that the server is up at an arbitrary moment, what is the probability that the current up period has lasted at most 12 hours? Compare with $P(Z \leq 12) = 0.2$ and explain the difference in one sentence. (Hint: on = “up and age ≤ 12 ”; the on-time per cycle is $\min(Z, 12)$.) (iv) Which of (i)–(iii) change if the repair time becomes deterministic (2 hours)? Which change if the up time becomes exp with mean 15 hours?

Part B — Critique an AI solution (find the flaws)

A student pasted the following into an AI assistant:

“At a bus stop, buses arrive with iid headways $\text{uniform}(10, 30)$ minutes and passengers arrive as a Poisson process with rate 3 per minute; every bus takes everyone waiting. The city pays the shelter operator 1 won per passenger-minute of waiting. (1) Mean waiting time per passenger and mean number waiting. (2) Long-run payment to the operator per minute. (3) Expected number of buses by $t = 15$ and by $t = 60$ minutes after the service starts (the first headway starts at $t = 0$). (4) The ticket machine at the stop works for an exp time with mean 50 hours and is then out of order for a $\text{uniform}(1, 3)$ -hour repair: long-run fraction of time it works.”

The AI replied:

Step 1. Headways have mean 20 min, so a passenger arriving at a random time sees on average half a headway ahead: $W = E[X]/2 = 10$ min.
 Step 2. By Little’s law, $L = \lambda W = 3 \times 10 = 30$ passengers waiting on average.
 Step 3. Payment per cycle: the number waiting grows at rate 3 per minute, so over a headway X the passenger-minutes are $3X^2/2$; with $E[X] = 20$ this is $3 \cdot 20^2/2 = 600$ won per cycle, and dividing by the mean headway gives $600/20 = 30$ won per minute.
 Step 4. The expected number of buses by t is $m(t) = t/E[X]$ (elementary renewal theorem): $m(15) = 0.75$ and $m(60) = 3$.
 Step 5. Check with Wald’s equation: $E[S_{N(t)}] = E[X]m(t)$. At $t = 60$ the last bus before 60 arrived on average around 50 minutes, so $m(60) \approx 50/20 = 2.5$, consistent with Step 4 up to rounding.
 Step 6. The availability formula $\lambda/(\lambda + \mu)$ requires exponential repair times. Since repairs are $\text{uniform}(1, 3)$, the long-run fraction cannot be computed from the means alone; approximating the repair by an exponential with mean 2 h gives $50/52 \approx 0.962$.

Your tasks:

- B1. Identify **at least three distinct errors** in this reasoning. Name each precisely (which quantity is confused with which; which limit is used as an identity; which hypothesis of which theorem is violated; which claim is false even though its number is right).

- B2. Write the correct solution: W and L (and the general formulas), the payment rate, $m(15)$ exactly and $m(60)$ (bounds and the asymptotic value on paper; the numerical value from HW4.ipynb), and the machine’s availability with the correct justification.
- B3. Verify W , L , the payment rate and $m(15)$, $m(60)$ by simulating the bus stop directly (a skeleton is provided; do not use any renewal formula inside the simulation). Report estimates with 95 % confidence intervals over independent replications.

Part C — Stress-test the assumptions (AI + simulation)

Two claims of this chapter deserve a test: “the rate of a renewal process depends only on the mean gap” (true, but how fast does $N(t)/t$ get there?) and “on/off fractions depend only on the means” (true for one machine — and in HW3 Part C it failed for two machines sharing a repair person).

- C1. **Convergence speed.** Take gaps with mean 1 and coefficient of variation $cv \in \{0, 0.5, 1, 2\}$ (deterministic, Erlang(4), exponential, hyperexponential or lognormal). *Before simulating*, discuss with your AI assistant and predict the time $t_{5\%}$ after which $|N(t)/t - 1| \leq 0.05$ holds with probability 95 % (hint: $N(t)$ is approximately normal with mean t/μ and variance $\sigma^2 t/\mu^3$). Then estimate $t_{5\%}$ by simulation over replications (skeleton provided) and compare. Which process would you rather estimate a rate from, and by how much does the required observation window grow with the cv ?
- C2. **Insensitivity, explained by cycles.** (a) Simulate one machine with up times of mean 2 days and repair times of mean 1 day for several shapes (deterministic, Erlang(4), exponential, lognormal) and confirm that the availability is $2/3$ in every case. State in one sentence which theorem makes this obvious. (b) Re-run the two-machine / one-repair-person system of HW3 (simulator provided) with deterministic repair times. Explain, using the vocabulary of regenerative and alternating renewal processes, why a single machine’s up/down sequence is *not* an alternating renewal process here, and why the joint system is nevertheless regenerative when the up times are exponential (name a regeneration epoch). Ask your AI assistant to state the general result behind (a); summarise it in at most five sentences.
- C3. (Optional) **The transient of the renewal function.** For the four gap distributions of C1 compute $m(t) - t/\mu$ with the grid solver and compare with $(\sigma^2 - \mu^2)/(2\mu^2)$. For deterministic gaps the difference does not converge — explain why with a sketch of $m(t)$, and relate it to the word *lattice* in Companion Proof 7.

Part D — Reflection and prompt log (keep it whenever AI was used)

Fill the table in HW4.ipynb (or reproduce it in your PDF): one row per meaningful AI interaction — where you used it, the prompt, what it returned, and **what you checked or corrected**. Then answer in 2–3 sentences: *where was the AI most useful, and where was it least trustworthy, on this homework?*

Logistics. This homework is not collected. Solutions to Part A will be posted on PLMS about 10 days after release. Parts B–D have no single “answer key”; bring your questions to class or office hours.